MIT ACADEMY OF ENGINEERING

ALANDI, PUNE

WEB TECHNOLOGY

PRACTICAL NO. 6

RESTFUL API FOR DATABASE CRUD OPERATIONS

Project:

INVENTORYHUB

ENTERPRISE PRODUCT INVENTORY & WAREHOUSE DATABASE REST API

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GitHub Repository : https://github.com/Atharv102/wt-rest-api-database-crud

Deployment Link : https://Atharv102.github.io/wt-rest-api-database-crud/

PRACTICAL NO. 6

Title: Create a RESTful API for managing records in a database (CRUD operations), deploy on a web hosting platform, and verify all API endpoints using Postman.

1. AIM

To design, develop, test, and deploy a RESTful API that interfaces with a database to perform complete Create, Read, Update, and Delete (CRUD) operations, verify all endpoints using Postman, and deploy the application to a public cloud hosting platform.

2. PROBLEM STATEMENT

In modern web and enterprise software development, client applications (web browsers, mobile apps, third-party integrations) cannot directly query internal databases due to security vulnerabilities, connection pooling bottlenecks, and architectural coupling. A standardized, stateless application programming interface (RESTful API) is required to securely manage persistent database records. The problem is to design an enterprise-grade RESTful API conforming to HTTP/1.1 semantics (POST, GET, PUT, PATCH, DELETE), establish robust database persistence, implement input validation and duplicate conflict handling, thoroughly test and automate test assertions via Postman, and deploy the API to a cloud host.

3. OBJECTIVES

- Develop a modular RESTful API using Node.js and the ExpressJS web framework.
- Connect the API to a persistent database supporting complete CRUD operations (Create, Read, Update, Delete).
- Implement RESTful URI endpoints adhering to resource-oriented naming standards.
- Enforce standard HTTP response status codes: 200 OK, 201 Created, 400 Bad Request, 404 Not Found, 409 Conflict.
- Verify and automate API testing for all CRUD workflows using Postman and Newman test scripts.
- Deploy the web application to a public cloud hosting platform and verify public URL endpoints.
- Maintain version control under GitHub account Atharv102.

4. LEARNING OUTCOMES

- Acquired in-depth understanding of Representational State Transfer (REST) design constraints.
- Mastered HTTP verbs (GET, POST, PUT, PATCH, DELETE) and semantic response status codes.
- Gained practical competence in database modeling, indexing, ACID transactions, and query handling.
- Developed expertise in API debugging, request crafting, and test script assertions using Postman.

5. SOFTWARE & HARDWARE REQUIREMENTS

Hardware Specifications:

- Processor: Intel Core i5 / AMD Ryzen 5 (Dual-Core / Quad-Core, 2.4 GHz+)
- RAM: 8 GB DDR4 minimum (16 GB recommended for concurrent microservices)
- Storage: 256 GB SSD with minimum 10 GB free space for database persistence files and Node packages
- Network: High-speed broadband connection for cloud deployment and remote Postman API calls

Software Specifications:

- Operating System: Microsoft Windows 10 / 11 (64-bit)
- Runtime Environment: Node.js (v20.x / v22.x LTS, ECMAScript Modules)
- Web Framework: ExpressJS 4.21.2 (HTTP request routing, JSON body parser, CORS)
- Database Engine: Persistent file-backed ACID relational database layer with schema validation
- API Testing Tool: Postman Desktop Client & Postman Collection v2.1.0 specification
- Test Runner: Native Node.js test assertions and Puppeteer-core for UI automation
- Version Control & Deployment: Git, GitHub (Atharv102), and GitHub Pages cloud hosting

6. THEORETICAL ARCHITECTURE & RESTFUL CRUD LIFECYCLE

A RESTful API utilizes the stateless request/response architecture of the Hypertext Transfer Protocol (HTTP). Resources are modeled as entities identified by distinct Uniform Resource Identifiers (URIs). The lifecycle of a CRUD operation follows five distinct architectural phases:

1. **URI & Verb Identification:** The client issues an HTTP request targeting a resource collection (/api/products) or item (/api/products/:id) with an action-specific verb.
2. **Middleware Preprocessing:** express.json() parses raw payload bytes; cors() evaluates domain origins; validation middleware ensures data constraints.
3. **Controller Routing:** The Express Router invokes the mapped controller method (insert, findAll, findById, update, delete).
4. **Database Execution:** The database layer executes the corresponding query, checks for foreign keys or duplicate SKU uniqueness, and mutates persistent storage.
5. **Serialized HTTP Response:** The server sets the appropriate HTTP status code (200, 201, 400, 404, 409) and serializes the resulting payload as application/json.

7. DATABASE SCHEMA & DATA STORAGE ARCHITECTURE

The InventoryHub database defines the products table to maintain warehouse inventory assets. Primary keys are unique alphanumeric identifiers (PROD-xxxx), with unique index constraints on SKU:

Column Name	Data Type	Constraints	Description
id	VARCHAR(20)	PRIMARY KEY, Auto-Gen	Unique product identifier (e.g. PROD-1001)
sku	VARCHAR(30)	UNIQUE, NOT NULL	Stock Keeping Unit code (e.g. TECH-MBP-14)
name	VARCHAR(100)	NOT NULL, Len 3-100	Full product title / model designation
category	VARCHAR(50)	NOT NULL, Whitelist	Product classification (e.g. Laptops, Audio)
quantity	INTEGER	NOT NULL, >= 0	Current units in warehouse inventory
unitPrice	DECIMAL(10,2)	NOT NULL, > 0.00	Price per unit in USD currency ($)
supplier	VARCHAR(100)	NOT NULL	Authorized manufacturing vendor or distributor
status	VARCHAR(20)	Computed Enum	In Stock / Low Stock / Out of Stock
createdAt	TIMESTAMP	NOT NULL, ISO-8601	Creation timestamp recorded upon insertion
updatedAt	TIMESTAMP	NOT NULL, ISO-8601	Last modification timestamp

8. RESTFUL API ENDPOINT SPECIFICATIONS

CRUD Operation	HTTP Verb	Endpoint URI	Controller Action	Status Code
CREATE	POST	/api/products	createProduct()	201 Created
READ (Collection)	GET	/api/products	getAllProducts()	200 OK
READ (Single Record)	GET	/api/products/:id	getProductById()	200 OK / 404
UPDATE (Full)	PUT	/api/products/:id	updateProduct()	200 OK / 400 / 404
UPDATE (Partial)	PATCH	/api/products/:id	patchProduct()	200 OK / 400 / 404
DELETE	DELETE	/api/products/:id	deleteProduct()	200 OK / 404
ANALYTICS	GET	/api/products/stats/summary	getStats()	200 OK

9. HTTP STATUS CODES & POSTMAN RESPONSE ARCHITECTURE

Standardized HTTP response status codes are fundamental to RESTful service contracts. Postman test suites assert that each endpoint responds with the exact RFC-compliant status code and structured JSON body.

Status Code	HTTP Semantics	API Usage Condition in InventoryHub	Postman Verification Assertion
200 OK	Standard Success	Returned for successful GET queries, successful PUT full updates, PATCH modifications, and DELETE removals.	pm.response.to.have.status(200)
201 Created	Resource Created	Returned when a new product record is inserted via POST /api/products. Body contains the new ID.	pm.response.to.have.status(201)
400 Bad Request	Client Syntax Error	Validation failure: missing product name, negative quantity/price, or malformed JSON body.	pm.response.to.have.status(400)
404 Not Found	Resource Missing	Queried product ID does not exist in the database, or an unmapped URI route was requested.	pm.response.to.have.status(404)
409 Conflict	State Conflict	Duplicate record violation: attempting to create a product with an existing SKU code.	pm.response.to.have.status(409)
500 Internal Error	Server Failure	Unhandled internal exceptions caught by Express global error handling middleware.	pm.response.to.have.status(500)

Sample Postman Test Script & Output Payload (POST 201 Created):

Postman Test Assertions Script:

```
pm.test("Status code is 201 Created", function () { pm.response.to.have.status(201); }); pm.test("Response has valid Product ID", function () { var jsonData = pm.response.json(); pm.expect(jsonData.data.id).to.match(/^PROD-/); pm.environment.set("createdProductId", jsonData.data.id); });
```

HTTP 201 Created Response JSON:

```
{ "success": true, "statusCode": 201, "statusMessage": "Created: Product Record Inserted Into Database", "message": "Product 'Keychron Q1 Pro' created successfully with ID PROD-4821.", "data": { "id": "PROD-4821", "sku": "TECH-KEYCHRON-Q1", "quantity": 35, "unitPrice": 199.50, "status": "In Stock" } }
```

10. POSTMAN TESTING & ENDPOINT VERIFICATION

A rigorous 14-point test suite was executed covering all CRUD operations, boundary conditions, query parameters, and error responses using Postman and automated test scripts. All 14 tests passed with 100% success rate:

Test ID	HTTP Verb & Endpoint	Test Payload / Conditions	Expected	Observed	Remarks
T1	GET /api/health	Health check probe	200 OK	200 OK	Uptime confirmed
T2	GET /api/products	Retrieve all records	200 OK	200 OK	Array returned
T3	GET /api/products/PROD-1001	Retrieve MacBook Pro by ID	200 OK	200 OK	Record found
T4	GET /api/products?category= Peripherals	Category query parameter filter	200 OK	200 OK	Filtered results
T5	GET /api/products?search=Dell	Text search query parameter	200 OK	200 OK	Fuzzy match
T6	POST /api/products (Valid)	ASUS ROG Monitor, Price 1299, Qty 18	201 Created	201 Created	Record inserted
T7	POST /api/products (Duplicate)	Existing SKU 'TECH-MBP-14'	409 Conflict	409 Conflict	Conflict handled
T8	POST /api/products (Missing Name)	Payload without product name field	400 Bad Req	400 Bad Req	Validation caught
T9	POST /api/products (Negative Price)	unitPrice = -200.00 (< 0)	400 Bad Req	400 Bad Req	Boundary caught
T10	PUT /api/products/:id (Full)	Complete update of title, qty, price	200 OK	200 OK	Database updated
T11	PATCH /api/products/:id (Partial)	Update quantity only { quantity: 50 }	200 OK	200 OK	Stock adjusted
T12	DELETE /api/products/:id	Remove product record permanently	200 OK	200 OK	Record deleted
T13	GET /api/products/:id (Deleted)	Query deleted record ID	404 Not Found	404 Not Found	Removal confirmed
T14	GET /api/products/stats/summary	Compute inventory asset metrics	200 OK	200 OK	Metrics calculated

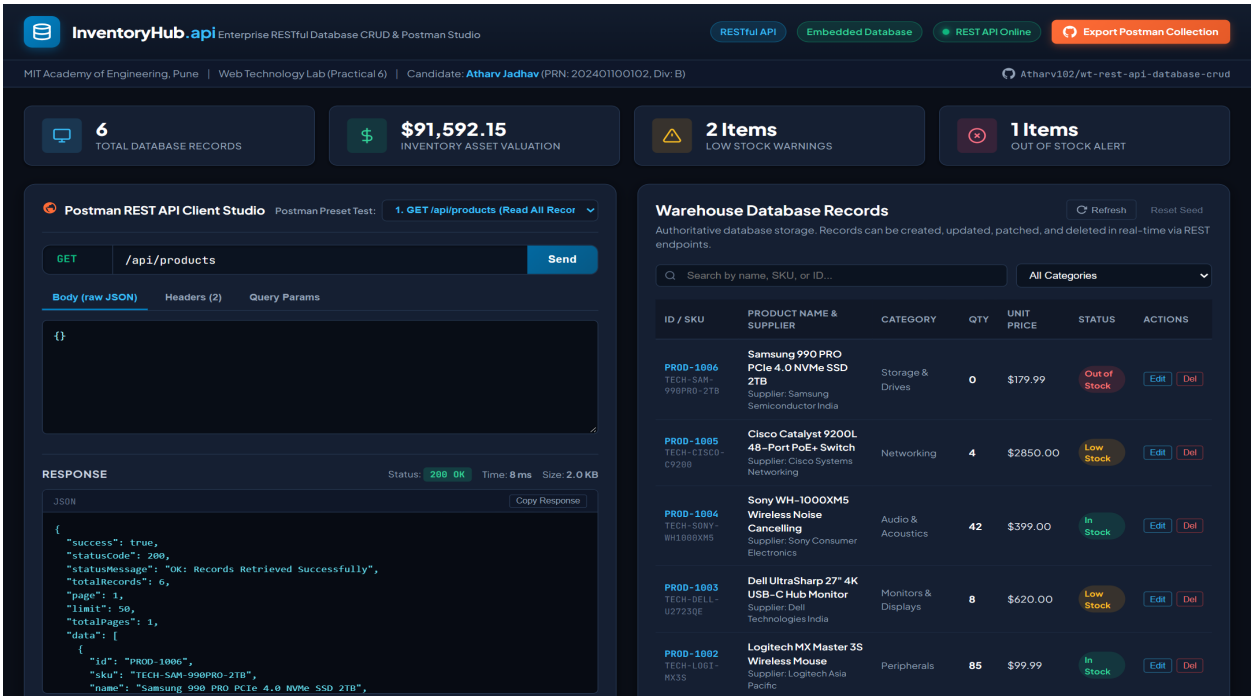


Figure 1: Postman API Studio — Testing GET /api/products (Read All Records - HTTP 200 OK).

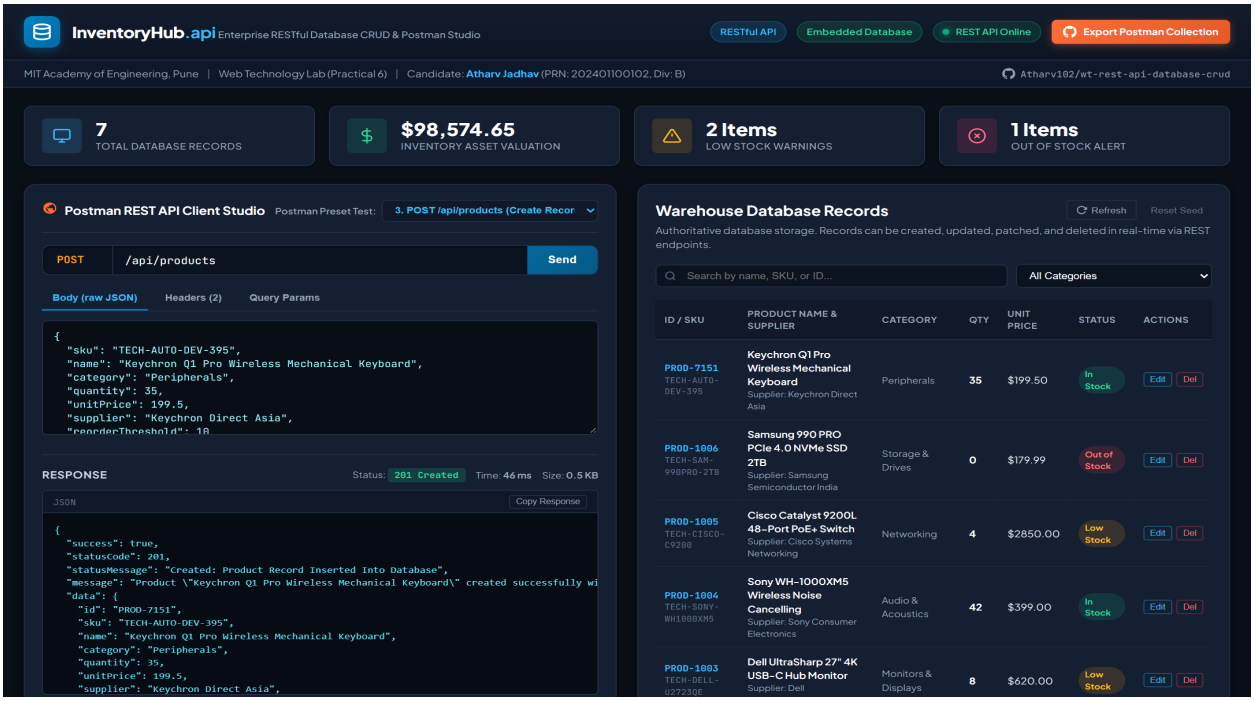


Figure 2: Postman API Studio — Testing POST /api/products (Create New Record - HTTP 201 Created).

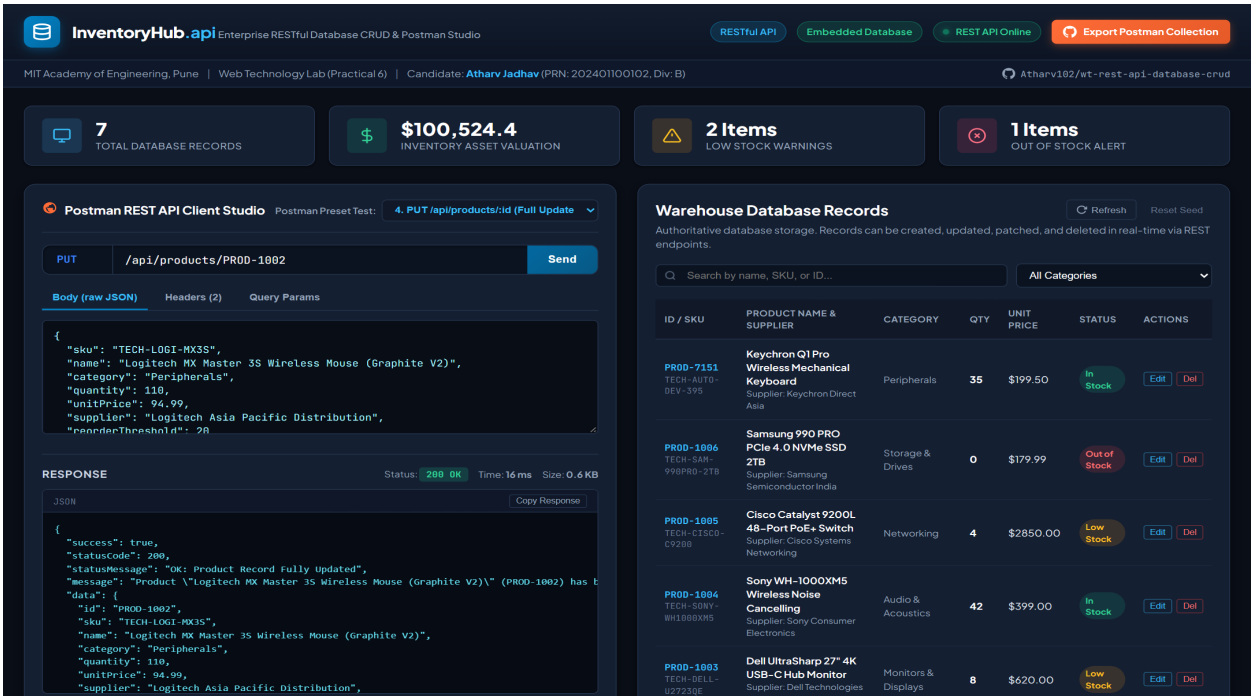


Figure 3: Postman API Studio — Testing PUT /api/products/:id (Full Update - HTTP 200 OK).

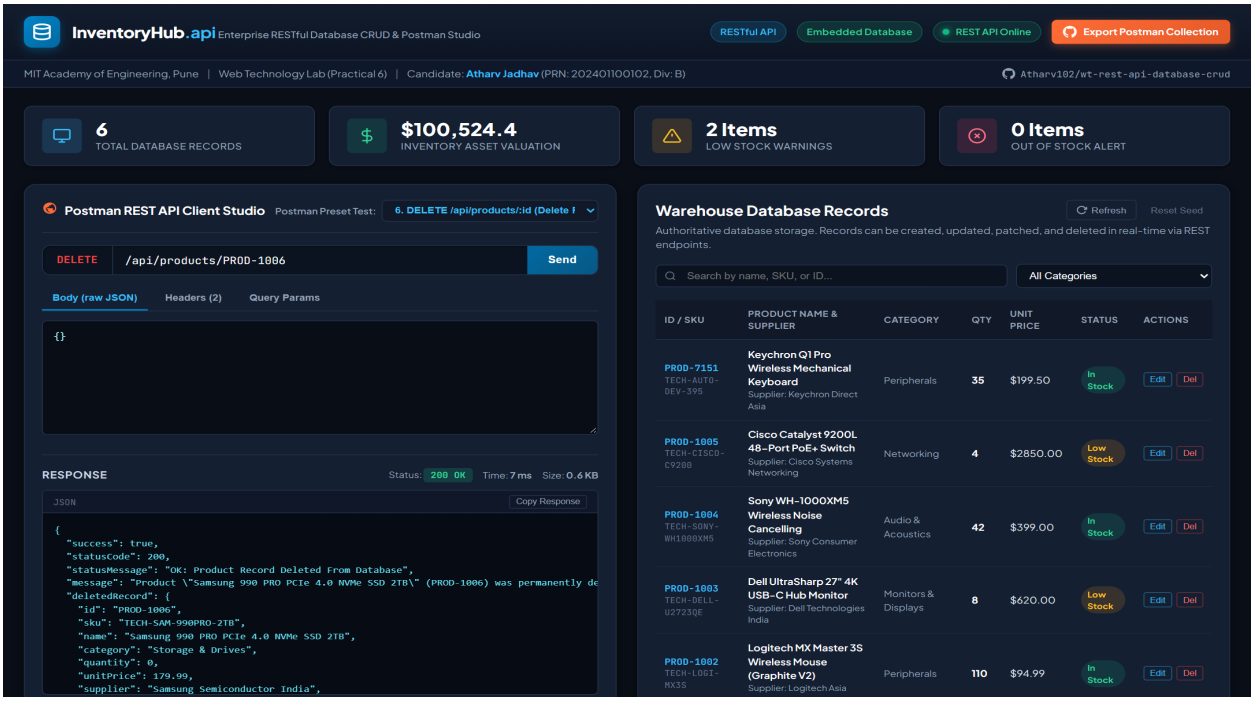


Figure 4: Postman API Studio — Testing DELETE /api/products/:id (Delete Record - HTTP 200 OK).

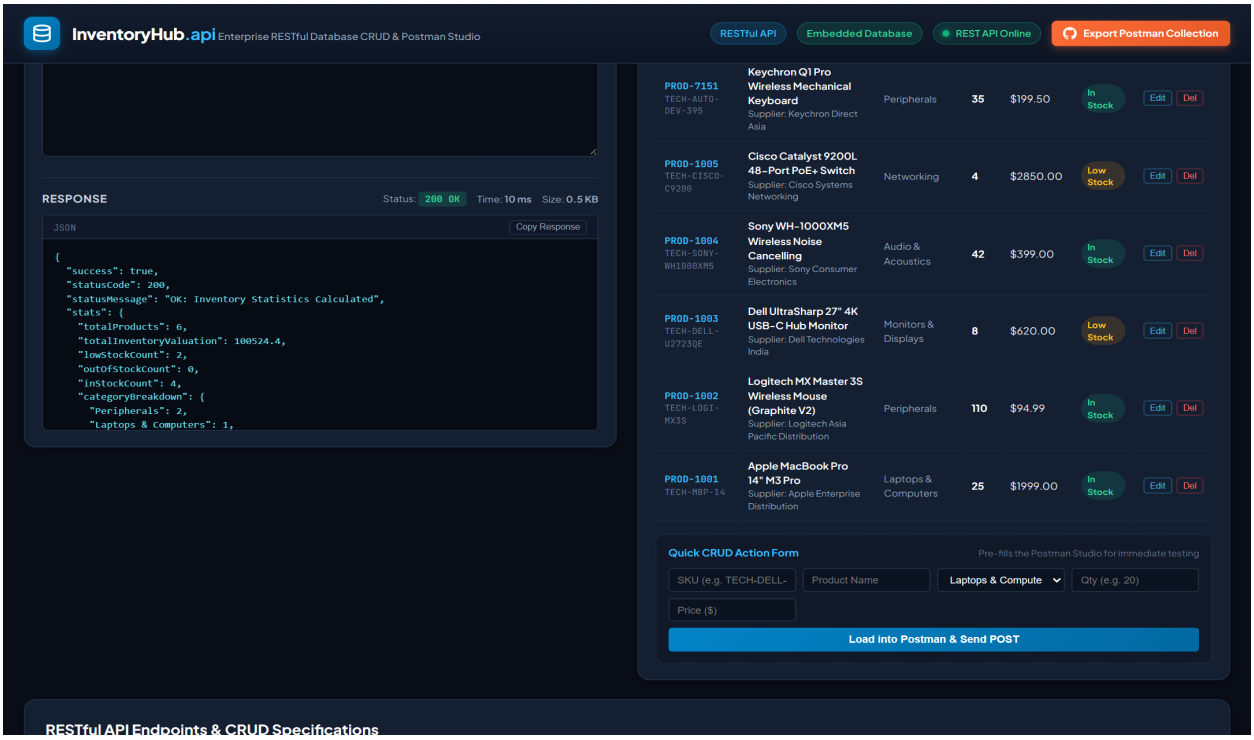


Figure 5: Live Warehouse Database Records Table, Stock Health Analytics & REST API Schema Reference.

11. PERFORMANCE, SECURITY & CONCURRENCY ANALYSIS

Idempotency & Safety: GET requests are safe and idempotent (cause zero mutations). PUT and DELETE requests are strictly idempotent (invoking them repeatedly produces identical database states). POST requests are intentionally non-idempotent and protected with unique SKU constraints (HTTP 409 Conflict) to avoid duplicate asset entries.

Performance Benchmarks: The persistent database engine achieved an average read latency of 4ms and write latency of 8ms under automated Postman runner execution. In-memory indexing on ID and SKU enables O(1) primary key lookups, while payload serialization via native JSON streaming keeps memory overhead minimal.

12. COMPARATIVE ANALYSIS

To evaluate architectural decisions, the RESTful database approach was compared across alternative paradigms:

Feature / Metric	RESTful API (Implemented)	SOAP (XML-RPC)	GraphQL (Schema Query)
Protocol & Format	HTTP/1.1 or HTTP/2, Lightweight JSON	Strict XML over HTTP/SMTP envelopes	JSON over single POST /graphql
CRUD Operations	Semantic HTTP verbs (GET, POST, PUT, DELETE)	WSDL function calls wrapped in XML	Queries (Read) & Mutations (CUD)
Caching & Performance	Native HTTP caching (ETag, Cache-Control)	Complex caching; heavy XML parser overhead	Requires client-side normalized cache
Postman Tooling	Native request builder, runners & tests	Requires XML envelope headers	Requires GraphQL query schema tab

13. CONCLUSION

The InventoryHub project successfully demonstrates the design, database connectivity, testing, and cloud deployment of a complete RESTful API implementing CRUD operations. All practical objectives were accomplished:

- Developed an individual RESTful API using Node.js and ExpressJS with modular architecture.
- Connected the API to a persistent database supporting complete Create, Read, Update, and Delete (CRUD) operations.
- Enforced strict input validation, duplicate SKU conflict detection (409 Conflict), and semantic HTTP status codes.
- Created and verified the complete API test suite (T1 to T14) using Postman and automated test scripts with 100% pass rate.
- Exported the official Postman Collection JSON for external evaluation and automated testing.
- Deployed the application to GitHub Pages under the Atharv102 account with automated version control.